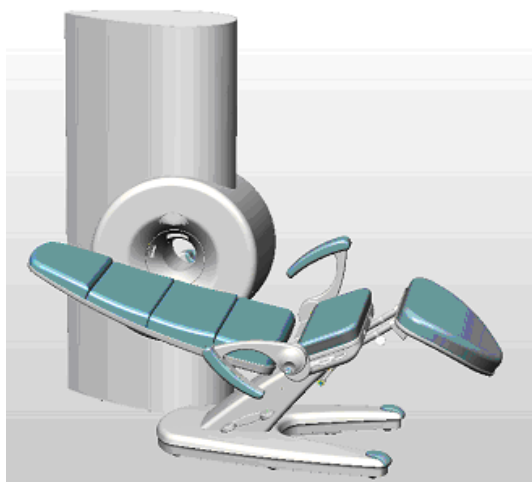


Field Replaceable Unit (FRU) Removal and Replacement Guide



Revision History

Revision	Date	Changes
1.0	January, 2011	Initial conversion to GE format
2.0	February, 2011	Update to exciter/receiver, gradient amplifier, and gradient coil timing per MRIhc 55105

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1 Overview

Scope

This guide describes methods for replacing field-replaceable units (FRUs) in the OPTIMA MR430S imaging system.

This guide includes the following topics:

- [Overview](#)- this chapter, which describes the *FRU Removal and Replacement Guide*.
- [Systems Cabinet](#) on page 7 describes procedures for replacing components of the systems cabinet.
- [Operator Console](#) on page 31 describes procedures for replacing components of the operator console.
- [Magnet Assembly](#) on page 33 describes procedures for replacing components of the magnet assembly.
- [Gradient Coil](#) on page 37 describes procedures for replacing components of the gradient coil.
- [Compressor](#) on page 47 describes procedures for replacing components of the compressor assembly.
- [Patient Chair](#) on page 49 describes procedures for replacing components of the patient chair.
- [Miscellaneous](#) on page 59 describes procedures for replacing components not covered in the other sections.

Hardware Requirements

Standard tools (such as screwdrivers, adjustable wrenches, and so on) are required. When other tools or materials are required, the procedures call them out. A manual lift ships with the system for replacing heavy items in the systems cabinet.

Service Philosophy

FRUs are defined as those components that can be replaced at the customer site. Some components require that the subsystem they are part of be returned to GE for service. The procedures for replacing some components are self-evident. Those procedures are not included in this guide.

General Service Notes

Aeroquip® Fittings

The OPTIMA MR430S system uses self-sealing Aeroquip fittings to connect gas transfer hoses and other gas-filled components. These fittings require some special care:

- When uncoupling two Aeroquip fittings, ***always use two wrenches***. If the friction between the two fittings is greater than the friction that secures one of the fittings to the component it is attached to, the fitting can unscrew from the component. If this happens, the results often generate unnecessary remedial work.
- Before coupling two Aeroquip fittings, verify that there is a sealing ring in the face of the male fitting, and that the sealing ring is not damaged. If the sealing ring is missing or damaged, the coupling will leak.

Testing

The tables in the Test part of each section list the test and calibration procedures required to qualify the system after replacement of the component. The tests and procedures are listed in the order they should be performed. Some tests or procedures are listed twice, for preparation then for final calibration.

Verify proper operation of all related functions after replacing any cable.

Reporting

In-process test data are not required to be submitted with the related Request for Service (RFS). However, it is ***mandatory*** that the system Daily Quality Assurance (DQA) test be run and passed, and results recorded on the RFS at the completion of the work.

Conventions

The conventions described below are used throughout the OPTIMA MR430S service documents.

Text Conventions

The following text conventions indicate on-screen features and controls, and emphasize important operations:

- References to on-screen text and controls are in a special font, to distinguish them from the normal flow of text.

Example: When Clone completed Successfully appears, remove the floppy and CD-ROM disks.

In the example, “Clone completed Successfully” is text in the computer display.

- When a reference to on-screen text includes an undefined variable, the variable is in italics, between brackets.

Example: No connection to Server <Orthone:[*database number*]>

In the example, “[*database number*]” refers to a value that is not specified. The rest of the example is the same as the display text.

- In some procedures, the most significant words in a step are in **bold type**. If the words refer to an on-screen feature, they are in the **special font in bold**.

Example: On the Start Copying Files window, click **Next**.

In the example, the N in “**Next**” is underlined because it is underlined in the display.

Terminology

The following terms have the meanings listed here:

- “Press” means to press a key or switch on a keyboard or other input device. (It does not refer to operating on-screen controls, so the name of the control is in the regular font.)

Example: Press **Enter**.

- “Click” means to place the mouse cursor on an item on the screen, and press the left mouse button.

Example: Click **Finish**.

Clicking causes the computer to perform an operation, such as opening or closing a window, saving data, or executing a task.

- “Double-click” means to place the mouse cursor on an item on the screen, and press the left mouse button twice.

Example: Double-click the **Network** icon.

- “Right-click” means to place the mouse cursor on an item on the screen, and press the right mouse button.

Example: Right-click **Onibc** and click **Open**.

- “Select” means to pick an option or file name from a list, by clicking it or rolling the mouse cursor over it.

Example: From the Edit menu, select **Copy**.

Selecting usually has no immediate result, and requires a subsequent click operation to implement the selection.

Safety Hazard Alerts

There are three hazard classifications used in this document:

Alert	Circumstance for Use
	Danger indicates an imminently hazardous situation, that, if not avoided, will result in death or serious injury. Danger is limited to extreme situations.
	Warning indicates a potentially hazardous situation, that, if not avoided, could result in death or serious injury.
	Caution indicates a potentially hazardous situation, that, if not avoided, may result in moderate to minor injury.

Electrical Safety Symbols



The MR system is a Class 1, type B applied part, rated for continuous operation. This device classification refers to the level of protection provided against electrical shock to the parts are in contact with the patient.



The symbol shown to the left, where displayed, indicates that there are dangerous voltages accessible beyond that point. Access may pose a safety risk and should only be done by qualified service personnel.



The symbol shown to the left indicates a safety warning and that accompanying documents should be consulted.



The symbol shown to the left indicates that this terminal is used for potential equalization.

NOTICE

Notes contain important information that can help you perform service operations. A note can describe a special condition you may encounter, can describe actions that simplify your work, or can alert you to other actions that create unnecessary work.

Required Personal Protective Equipment (PPE)

- Safety glasses
- Appropriate material-handling gloves
- Non-ferrous safety shoes

Certain procedures will also require the following:

- Cryogenic gloves
- Torso/Extremity protection
- Personal O₂ monitor

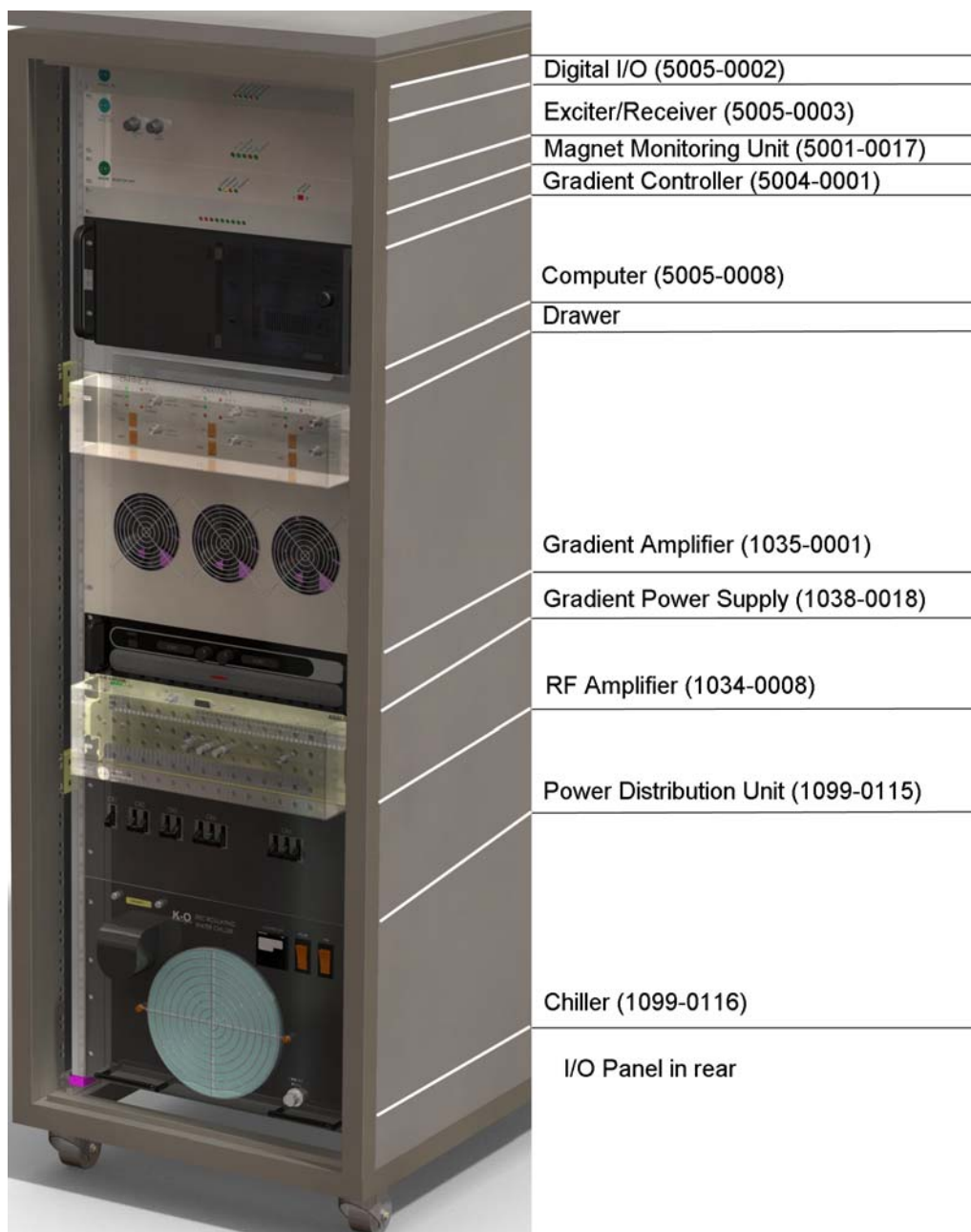
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2 Systems Cabinet

This section includes the following procedures:

- [Preparing the Systems Cabinet for Service](#) on page 8
- [Power Distribution Unit \(PDU\)](#) on page 8
- [I/O Panel](#) on page 10
- [RF Power Amplifier \(RF Amp\)](#) on page 11
- [Gradient Amplifier \(Gradient Amp\)](#) on page 12
- [Gradient Amp Power Supply](#) on page 20
- [Exciter-Receiver](#) on page 22
- [Digital I/O Module](#) on page 24
- [Magnet Monitoring Unit \(MMU\)](#) on page 25
- [Host Computer](#) on page 26
- [Chiller](#) on page 27

Figure 2-1 Systems Cabinet Assembly



Preparing the Systems Cabinet for Service

Before removing any components of the systems cabinet, you must complete the following procedure:

1. Perform the LOTO procedure for the systems cabinet.
2. Unlock the **rear door** of the systems cabinet, and open the door.

⚠ WARNING *When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.*

Power Distribution Unit (PDU)

Required Persons	Estimated Procedure Completion Time
2	2 hours

This procedure describes how to replace the power distribution unit (PDU) in the OPTIMA MR430S.

⚠ WARNING *Possible personal injury! Heavy Object - The PDU weighs approximately 185 pounds/84 Kg. Lifting this product requires assistance (mechanical or additional personnel).*

To replace the 1099-0115 PDU:

1. Complete the **procedure** described in [Preparing the Systems Cabinet for Service](#) on page 8.
2. Disconnect all the power cables from the outlets on the rear of the PDU (see [Figure 2-2](#)).

Figure 2-2 Rear of the PDU



⚠ CAUTION *To prevent damage to components, note which outlet on the PDU each power cable connects to. Connecting a cable that belongs in a Control Electronics outlet to a Power Electronics outlet can damage components.*

3. Disconnect the ground cable from the stud on the rear of the PDU.
4. Disconnect the cable labeled Power Control from the rear of the PDU.
5. Remove the four screws that secure the front of the PDU to the front of the systems cabinet.
6. Ensure that there's enough slack for the main power cable (CB4) and the gradient power supply power cable (CB4) to allow the PDU to slide out the front of the systems cabinet.

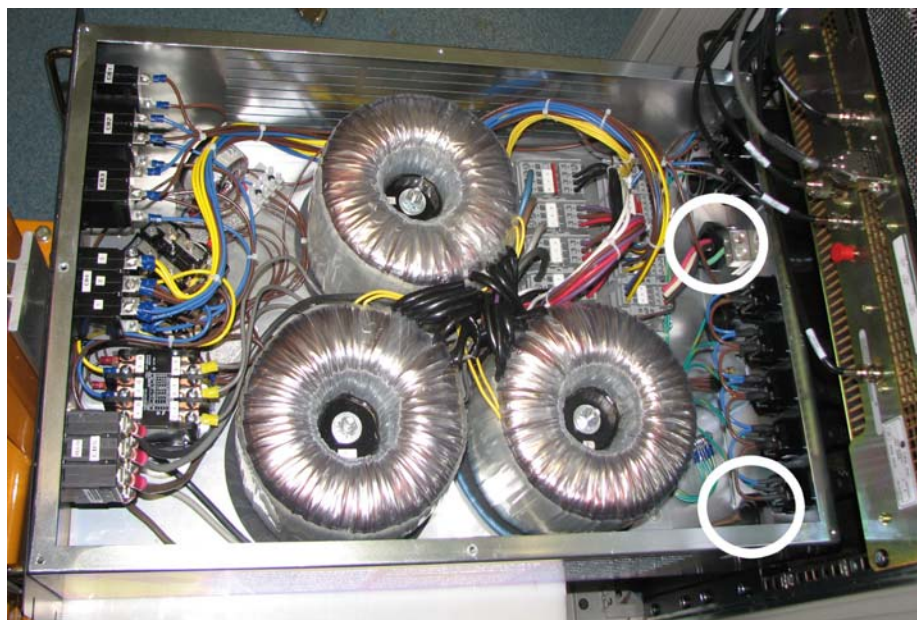
NOTICE *If there isn't enough slack for the gradient power supply power cable, disconnect it from the gradient power supply terminal block at the rear of the systems cabinet. Make sure to also disconnect the ground cable at the rear of the gradient power supply (see [Figure 2-3](#)).*

Figure 2-3 Gradient Power Supply Terminal Block



7. Ready and secure the Manual Lift in front of the PDU by lifting the platform up to the bottom height of the PDU and applying the caster brakes.
8. Slide the PDU out the front of the systems cabinet.
9. Rest the PDU on the Manual Lift device.
10. Remove the screws that secure the top cover of the PDU.
11. Disconnect the main power cable and the gradient power supply power cable from within the PDU. If needed, remove the two screws securing the main breaker to the front panel to ease access to cable connections (see [Figure 2-4](#)).

Figure 2-4 Top of PDU



12. Reverse the preceding steps to install the new PDU.
13. Remove LOTO for the systems cabinet.

Testing the PDU

Perform these tests with the power switches of all system cabinet components on, and all circuit breakers on the front of the PDU in the On position.

1. Verify that when the system is in Full On mode, all components are powered on.
2. Verify that when the system is in Standby mode, only the scanner computer, Digital IO, Exciter/Receiver, and MMU are powered on.
3. Log out of the applications software before executing the next step.
4. Verify that when the system is in Emergency Stop mode, all units in the cabinet are powered off. After an Emergency Stop you must go into Full on Mode to turn the system back on.

I/O Panel

Required Persons	Estimated Procedure Completion Time
1	2 hours

Replacing the I/O Panel

This procedure describes how to replace the I/O Panel in the Optima MR430s system (see [Figure 2-5](#)).

Figure 2-5 I/O Panel

To replace the 5005-0007 I/O panel:

1. Switch the compressor remote drive switch to INT (local mode).
2. Log into the scanner application software and enter Service Mode.
3. Click Service Tools, then click Free CAS.
4. Start CasWin.exe from the Service Tools pull-down menu.
5. Click the MMU1.5 tab
6. Deselect Enable Callout
7. Complete the procedure described in the [Preparing the Systems Cabinet for Service](#) on page 8.
8. Disconnect all cables connected to the I/O Panel.
9. Remove the four screws that secure the I/O Panel to the rear of the systems cabinet.
10. Replace the I/O Panel with the new one.
11. Reconnect all cables to the new I/O Panel.
12. Remove LOTO for the systems cabinet.



Testing the I/O Panel

After servicing or replacing the I/O panel, perform the tests listed in the following table.

Table 2-1: I/O Panel Tests

Test Description	Reference Document	Document #	Section
Monitor	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test
Modem Dial-in	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test
DICOM Transfer	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test
Magnet Monitoring Unit (MMU)	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test
Quench	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	3 - System Installation / Cooling Down the Magnet
PDU	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Miscellaneous Tests
DQA	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Characterizing the 180mm RF coil

RF Power Amplifier (RF Amp)

Required Persons	Estimated Procedure Completion Time
2	2 hours

Replacing the RF Amp

See [Figure 2-1](#) on page 7 to identify the RF power amplifier.

Tools and Test Equipment	Quantity	Part Number
RF Power Measurement Kit	1	46-317724G2
Oscilloscope (Min 100 MHz)	1	



WARNING When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8 before performing any service procedure inside the systems cabinet.



WARNING Possible personal injury! Heavy Object - The RF Amplifier with box weighs approximately 90 pounds/41 Kg. Lifting this product requires assistance (mechanical or additional personnel).

To replace the 1034-0008 RF amp:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Disconnect the power cable from the rear of the RF Amp. Make sure to also disconnect the ground cable from the stud on the rear of the RF Amp (see [Figure 2-6](#)).
3. Remove the protective cover from the front of the RF Amp.
4. Disconnect the following cables at the front of the RF Amp (see [Figure 2-7](#)):
 - a. Gating Input
 - b. Test Interface
 - c. RF out
 - d. Forward Monitor
 - e. RF Input

Figure 2-6 Rear of RF Amplifier

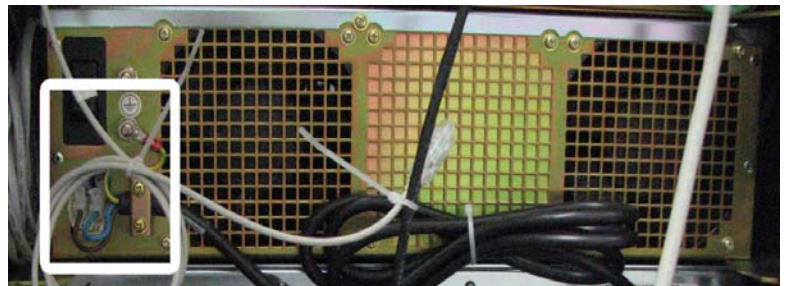


Figure 2-7 Front of the RF Amplifier



5. Remove the four screws that secure the front of the RF Amp to the systems cabinet.
6. Ready and secure the Manual Lift in front of the RF Amp by lifting the platform up to the bottom height of the RF Amp and applying the caster brakes.

7. Slide the RF Amp out the front of the systems cabinet.
8. Rest the RF Amp on the Manual Lift device.
9. To install a replacement RF Amp, reverse the preceding steps.
10. Remove LOTO for the Systems Cabinet.
11. In the ConfigFileEditor.exe file, update the RfAmp_Trip_Pt_Offset and RfAmp_Trip_Pt_Slope parameters to the values recorded on the back of the new RF Amplifier.

Testing the RF Amp

After replacing the RF amplifier, perform the tests listed in the following table.

Table 2-2: RF Amp Tests

Test Description	Reference Document	Document #	Section
TX Calibration	Transmitter Calibration	DOC0797559	--
External Loopback	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test/ RF Loopback Testing
180mm RF coil SNR	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil

Gradient Amplifier (Gradient Amp)

Required Persons	Estimated Procedure Completion Time
2	5 hours

Replacing the Gradient Amplifier

This procedure describes the removal and replacement of the Gradient Amplifier component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the gradient amplifier.

Tools and Test Equipment	Quantity	Part Number
Small Sample Kit	1	2000-9406



WARNING

When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.



WARNING

Possible personal injury! Heavy Object - The Gradient Amplifier with box weighs approximately 125 pounds/41 Kg. Lifting this product requires assistance (mechanical or additional personnel).

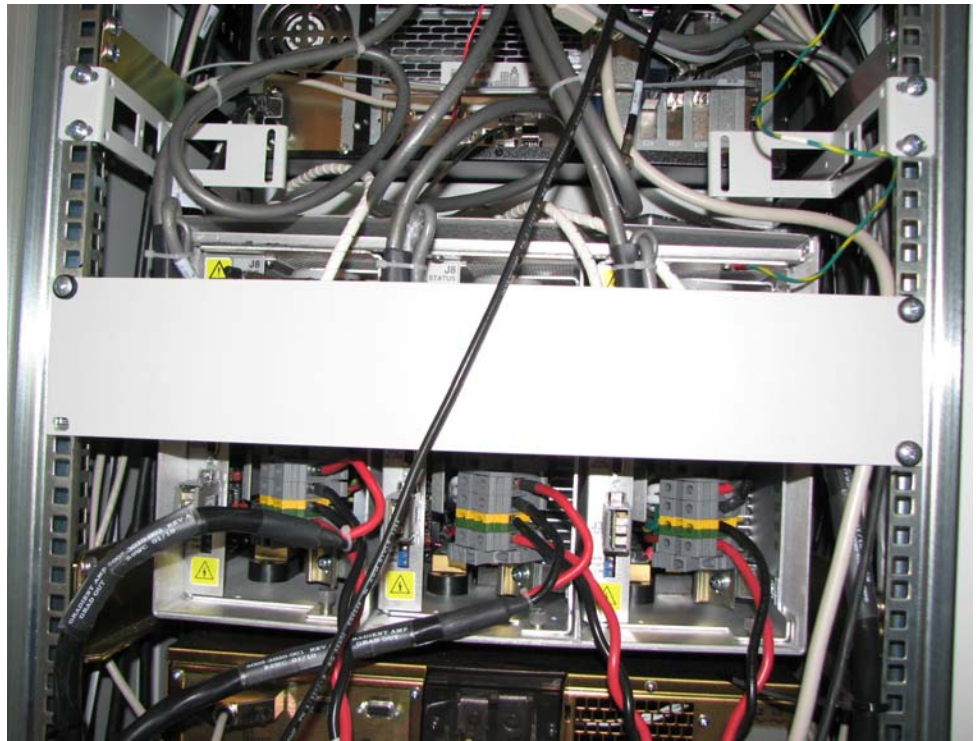
To replace the 1035-0001 Gradient Amplifier:

NOTICE

Remove the J3 connector first to prevent damaging the connector when removing other cables or devices.

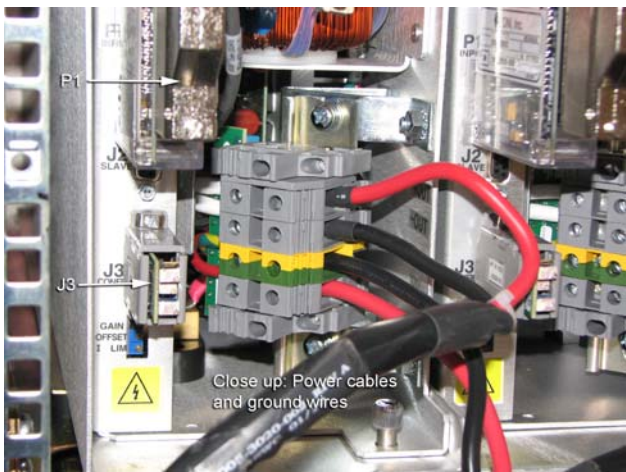
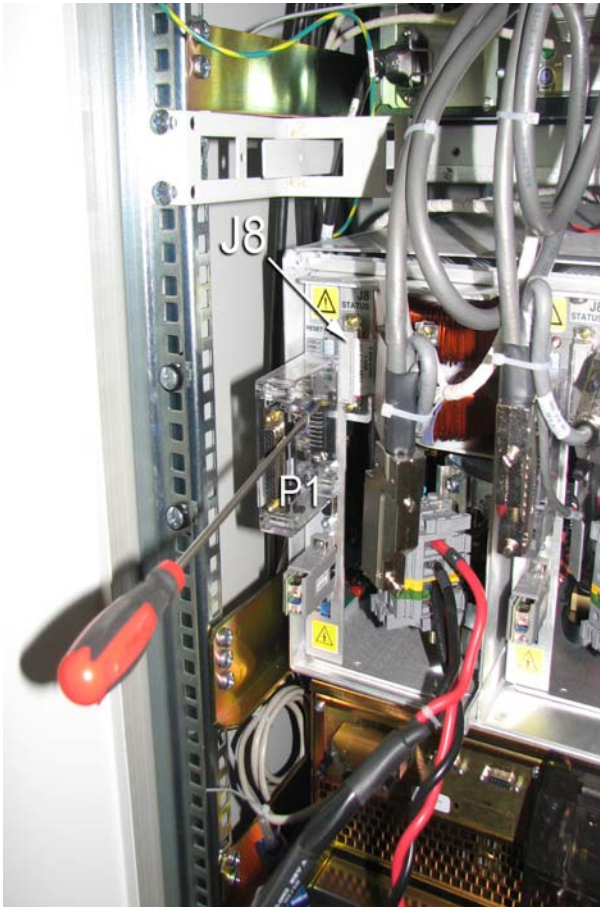
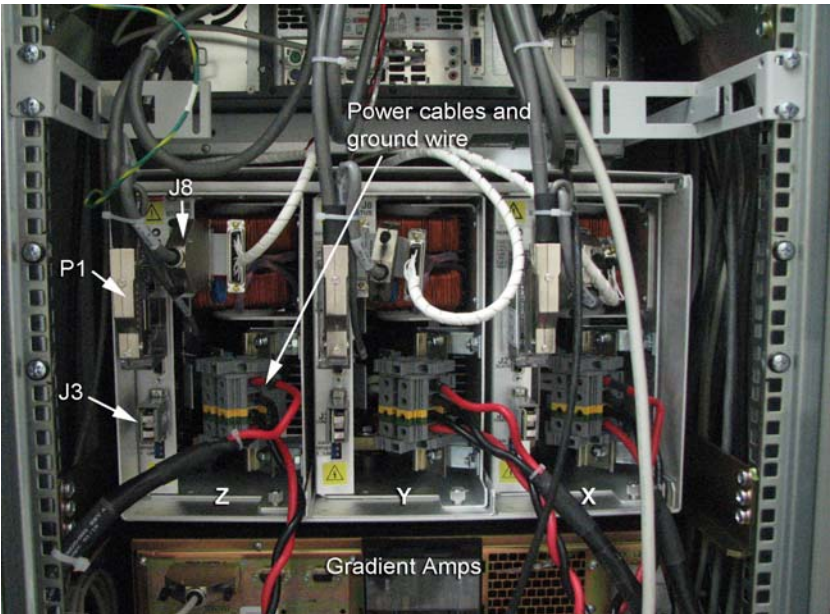
1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the cross plate to the back of the Systems Cabinet (see [Figure 2-8](#)).

Figure 2-8 System Cabinet Cross Plate



3. Disconnect the following cables and/or connectors from all 3 Gradient Amp Modules at the rear of the Gradient Amplifier component (see [Figure 2-9](#)):
 - a. J3, J8, P1
 - b. Power cables and Ground wire
 - c. Connector to PCB of all 3 gradient amp modules

Figure 2-9 Gradient Amplifier Module Removal



- Remove the 2 screws that secure front plastic cover and then remove the remaining 2 screws that secure the front of the Gradient Amplifier to the Systems Cabinet (see [Figure 2-10](#)).

Figure 2-10 Gradient Amplifier

- Pull the Gradient Amplifier out from the front of the Cabinet about 4"(100mm) to access the connections for the cabinet fan. Mark the connector plugs (for proper reconnection), then disconnect these plugs as well (see [Figure 2-11](#)).

Figure 2-11 Gradient Amplifier Connection Plugs

- Ready and secure the Manual Lift in front of the Gradient Amplifier by lifting the platform up to the bottom height of the Gradient Amplifier and applying the caster brakes.

Figure 2-12 Positioning the Manual Lift

7. Slide out the Amp from the front of the Cabinet and onto the Manual Lift platform. Before fully removing the Amplifier from the Cabinet, place a pad or block of about 1¼" thickness, if available, between the far end of the Lift platform and the bottom of the Amplifier. Finish sliding the Amplifier onto the platform, unlock the caster brakes, and move Lift and Gradient Amplifier out of the way (see [Figure 2-12](#) and [Figure 2-13](#)).

Figure 2-13 Removing the Gradient Amplifier

8. A Gradient Amplifier Module may be replaced at this time or follow the Gradient Amplifier - X, Y, and/or Z Module Replacement procedure. Secure Amplifier to platform when doing any work on the Amplifier in this position.
9. To install a replacement Gradient Amplifier, set a new one on the platform (with the optional pad or block in place) and reverse the preceding steps. Be sure to reconnect the fan connections in front prior to fully placing the Gradient Amplifier into the Systems Cabinet.
10. Reattach the cross plate at the rear of the Systems Cabinet.
11. Remove LOTO from the Systems Cabinet.
12. Perform the tests listed in the Table 2-3, "Gradient Amp Tests," on page 18.

Gradient Amplifier -- X, Y, and/or Z Module Replacement

Required Persons	Estimated Procedure Completion Time
1	4.5 hours

This procedure describes the removal and replacement of Gradient Amplifier Modules in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the gradient amplifier.

Tools and Test Equipment	Quantity	Part Number
Small Sample Kit	1	2000-9406

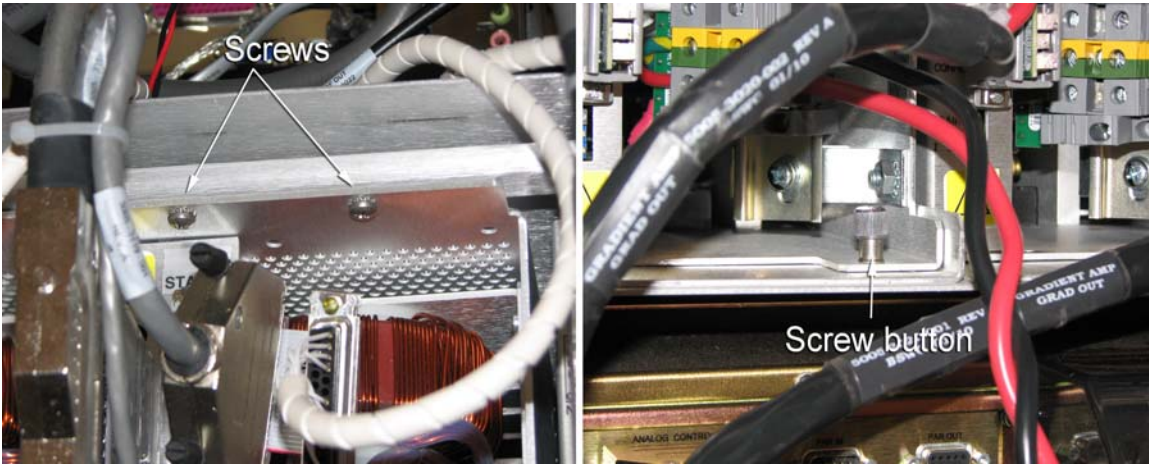
**WARNING**

When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

To replace the 1035-0001-001 Gradient Amplifier Module:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the cross plate to the back of the Systems Cabinet (see [Figure 2-8](#)).
3. Disconnect the following cables and/or connectors from the gradient amp module to be replaced at the rear of the Gradient Amplifier component (see [Figure 2-9](#)):
 - a. J3, J8, P1
 - b. Power cables and Ground wire
 - c. Connector to PCB
4. Remove the 2 screws (at the top rear of each module) that secure the module to the cross-bracket. Loosen the screw button on the bottom right of each module (see [Figure 2-14](#)).

Figure 2-14 Module and Cross Bracket



5. Slide the Gradient Amplifier Module out from the rear of the Gradient Amplifier Case. You will encounter interference to the cross-bracket from 2 screws attached to the top of the module (see [Figure 2-15](#)). Use a screwdriver between the module and the bracket to bend the bracket slightly upwards to enable the screw heads to pass underneath the bracket. Once clear of the interference, remove the module completely.

Figure 2-15 Removed Module



6. Replace the X, Y, and/or Z Gradient Amplifier Module(s) by sliding it into the appropriate slot(s) of the Gradient Amplifier Case. Reverse the above instructions to complete the removal and replacement.
7. Reattach the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

Testing the Gradient Amplifier

After servicing the gradient amplifier, perform the tests listed in the following table.

Table 2-3: Gradient Amp Tests

Test Description	Reference Document	Document #	Section
Gradient Calibration	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Gradient Amplifier Calibration
Eddy Current Compensation	Optima 1.5T MR430s Eddy Compensation Guide	DOC0797560	--
180mm RF coil SNR	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil
Image Quality/Ghosting	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Characterizing Image Quality

D/A Converter Daughterboard Replacement

Required Persons	Estimated Procedure Completion Time
1	1 hour

This procedure describes the removal and replacement of the D/A converter daughterboard in the Optima 1.5T MR430s Systems Cabinet.

Tools and Test Equipment	Quantity	Part Number
No additional tools required		



WARNING When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

To replace the 5004-0003 D/A converter daughterboard:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the cross plate to the back of the Systems Cabinet (see [Figure 2-8](#)).
3. Remove the cable from the back of the daughterboard.
4. Unscrew the daughterboard from the P1 connector. See [Figure 2-16](#) to see how to insert the screwdriver to access the P1 connector screws.
5. Remove the daughterboard by pulling the card straight back. Be careful not to twist or bend the card.
6. Install the replacement daughterboard by inserting the card straight in. Be careful not to twist or bend the card during insertion.
7. Reattach the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

Figure 2-16 D/A Converter Daughterboard



Testing the D/A Converter Daughterboard

After servicing the gradient amplifier, perform the tests listed in the following table.

Table 2-4: Gradient Amp Tests

Test Description	Reference Document	Document #	Section
Gradient Calibration	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test/ Gradient Amplifier Calibration
180mm RF coil SNR	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil

Gradient Controller Component Replacement

Required Persons	Estimated Procedure Completion Time
1	1.5 hours

This procedure describes the removal and replacement of the Gradient Controller component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the gradient controller.

⚠ WARNING *When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.*

To replace the 5004-0001 gradient controller:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the plate across the back of the Systems Cabinet (see [Figure 2-8](#)).
3. Remove the 2 screws that secure the fan box in the rear of the Systems Cabinet. Move fan out of the way (on top of cabinet, see [Figure 2-20](#)).
4. Disconnect the following cables at the rear of the Gradient Controller component (see [Figure 2-24](#)):
 - a. J1, J2, J3, J4, J5, J6, J7, J8
 - b. Power cable
5. Remove the 4 screws that secure the front of the Gradient Controller to the Systems Cabinet. Slide the component out the front of the Cabinet (see [Figure 2-22](#)).
6. To install a replacement Gradient Controller, reverse the preceding steps, and make sure the power switch at the rear of the Gradient Controller is switched ON.
7. Reattach the fan box and the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

Testing the Gradient Controller

After servicing the gradient controller, perform the tests listed in the following table.

Table 2-5: Gradient Controller Tests

Test Description	Reference Document	Document #	Section
180mm RF coil SNR	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil

Gradient Amp Power Supply

Gradient Amplifier Power Supply Replacement

Required Persons	Estimated Procedure Completion Time
1	1 hour

This procedure describes the removal and replacement of the Gradient Amplifier Power Supply component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the gradient amplifier.

⚠ WARNING *When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.*

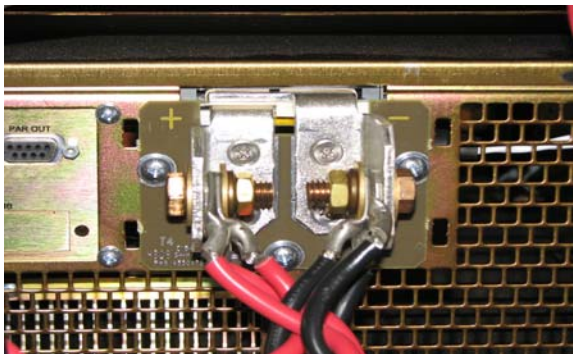
⚠ WARNING *Possible personal injury! Heavy Object - The Gradient Power Supply with box weighs approximately 62 pounds/28 Kg. Lifting this product requires assistance (mechanical or additional personnel).*

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Disconnect the following from the rear of the Gradient Amplifier Power Supply component:
 - a. Plastic protective cover (press sides outwards to disengage tabs) [2 each side, 4 total]
 - b. All power cables, the ground wire, and the Analog Control connector.

Figure 2-17 Rear of Gradient Amplifier Power Supply



Figure 2-18 Power Cable Connections



NOTICE

RED power leads go to (+) positive terminal (on the left) and the BLACK leads go to the (-) negative terminal (on the right).

3. Remove the 4 screws that secure the front of the Gradient Amplifier Power Supply to the Systems Cabinet.

Figure 2-19 Gradient Amplifier Power Supply

4. Ready and secure the Manual Lift in front of the Gradient Power Supply by lifting the platform up to the bottom height of the Gradient Power Supply and applying the caster brakes.
5. Slide the Gradient Amp Power Supply out from the front of the Cabinet.
6. To install a replacement Gradient Amplifier Power Supply, reverse the preceding steps and make sure the power switch on the front is switched ON.
7. Remove LOTO from the Systems Cabinet.



Testing the Gradient Amp Power Supply

After replacing the gradient amp power supply, perform the procedures listed in the following table.

Table 2-6: Gradient Amp Power Supply Tests

Test Description	Reference Document	Document #	Section
180mm RF coil SNR	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil

Exciter-Receiver

Required Persons	Estimated Procedure Completion Time
1	3 hours

Replacing the Exciter-Receiver

This procedure describes the removal and replacement of the Exciter-Receiver component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the exciter-receiver.

Tools and Test Equipment	Quantity	Part Number
RF Power Measurement Kit	1	46-317724G2
Oscilloscope (Min 100 MHz)	1	



WARNING When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

To replace the exciter-receiver:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the plate across the back of the Systems Cabinet (see [Figure 2-20](#)).

Figure 2-20 System Cabinet Plate and Fan Box

3. Remove the 2 screws that secure the fan box in the rear of the Systems Cabinet. Move fan out of the way (on top of cabinet).
4. Disconnect the following cables at the rear of the Exciter-Receiver component:
 - a. J1, J2, J3, J4, J7, J8, J9, J10
 - b. Power cable
5. Remove the 4 screws that secure the front of the Exciter-Receiver to the Systems Cabinet. Slide the component out the front of the Cabinet.
6. To install a replacement Exciter-Receiver, reverse the preceding steps, and make sure the power switch at the rear of the Exciter-Receiver is switched ON.
7. Reattach the fan box and the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

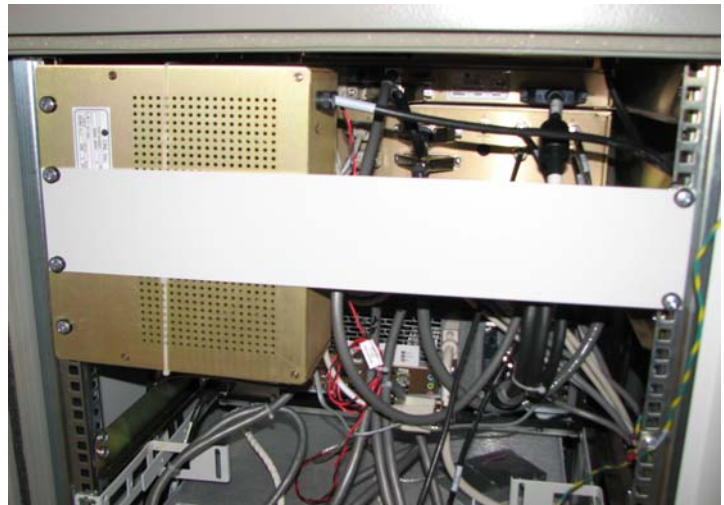


Figure 2-21 Rear of Exciter-Receiver



Figure 2-22 Modules at the Top of the Systems Cabinet



Testing the Exciter-Receiver

After servicing the exciter-receiver, perform the tests listed in the following table.

Table 2-7: Exciter-Receiver Tests

Test Description	Reference Document	Document #	Section
TX Calibration	Transmitter Calibration	DOC0797559	--
External Loopback	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test
180mm RF coil SNR	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Characterizing the 180mm RF Coil
123mm RF coil SNR	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Characterizing the 123mm RF Coil
RFI	Optima 1.5T MR430s System Installation Guide	DOC0763285	4 - System Configuration and Test/ Miscellaneous Tests - RF Interference

Digital I/O Module

Required Persons	Estimated Procedure Completion Time
1	1 hour

Replacing the Digital I/O Module

This procedure describes the removal and replacement of the Digital I/O component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the digital I/O module.

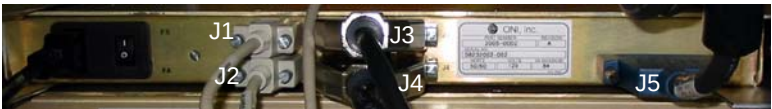
**WARNING**

When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

To replace the 5005-0002 digital I/O module:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the plate across the back of the Systems Cabinet (see [Figure 2-20](#)).
3. Remove the 2 screws that secure the fan box in the rear of the Systems Cabinet. Move fan out of the way (on top of cabinet).
4. Disconnect the following cables at the rear of the Digital I/O component:
 - a. J1, J2, J3, J4, J5
 - b. Power cable
5. Remove the 4 screws that secure the front of the Digital I/O to the Systems Cabinet. Slide the component out the front of the Cabinet (see [Figure 2-22](#)).
6. To install a replacement Digital I/O, reverse the preceding steps, and make sure the power switch at the rear of the Digital I/O is switched ON.
7. Reattach the fan box and the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

Figure 2-23 Rear of Digital I/O Module



Testing the Digital I/O Module

After servicing the digital I/O module, perform DQA scan. See [OPTIMA 1.5T MR430s System Installation Guide](#), Section 4, System Configuration and Test/ Characterizing the 180mm RF Coil.

Magnet Monitoring Unit (MMU)

Required Persons	Estimated Procedure Completion Time
1	1 hour

Replacing the MMU

This procedure describes the removal and replacement of the Magnet Monitoring Unit (MMU) component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to locate the MMU.

WARNING When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

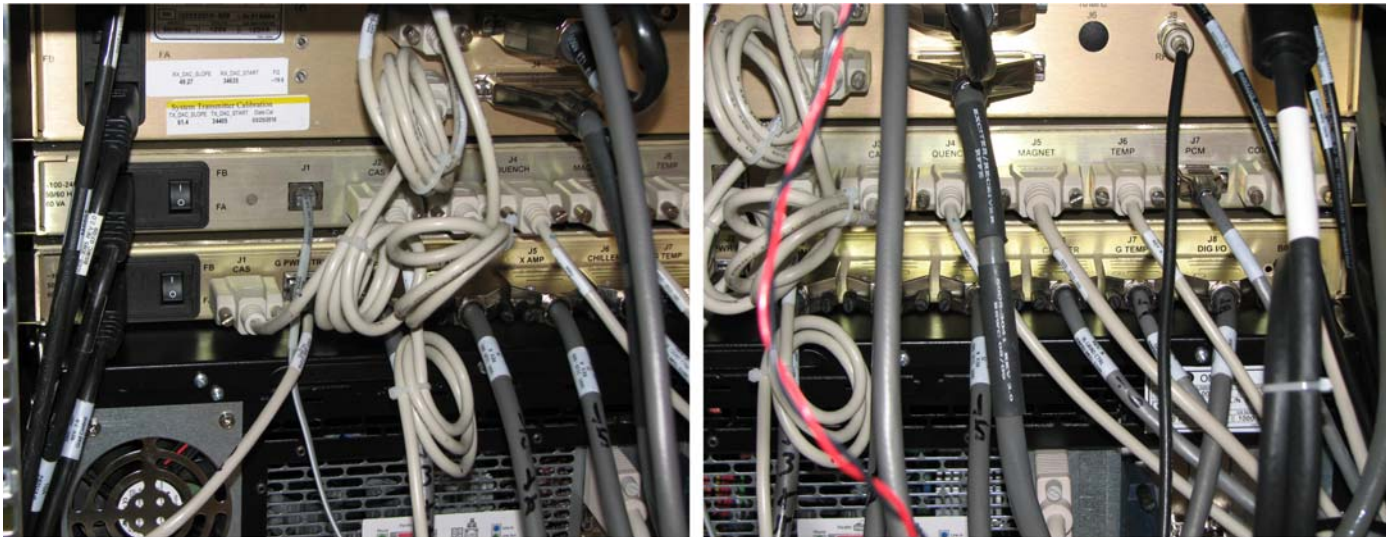
NOTICE Minimize the amount of time the Systems cabinet is off as the magnet pressure regulation will not be active while MMU cables are disconnected.

To replace the 5001-0017 MMU:

1. Complete the procedure [Preparing the Systems Cabinet for Service](#) on page 8.
2. Remove the 4 screws that secure the plate across the back of the Systems Cabinet (see [Figure 2-20](#)).
3. Remove the 2 screws that secure the fan box in the rear of the Systems Cabinet. Move fan out of the way (on top of cabinet).
4. Disconnect the following cables at the rear of the MMU component:
 - a. J1, J2, J3, J4, J5, J6, J7, J8
 - b. Power cable

WARNING Removing the J8 cable will interrupt the signal to the MMU and shut down the Cryogenic Compressor. Switch the compressor remote-drive switch to INT (local mode) or keep the system downtime to a minimum and reconnect this cable as soon as possible.

Figure 2-24 Rear of MMU



5. Remove the 4 screws that secure the front of the MMU to the Systems Cabinet. Slide the component out the front of the Cabinet (see [Figure 2-22](#)).
6. To install a replacement MMU, reverse the preceding steps, and make sure the power switch at the rear of the MMU is switched ON.
7. Reattach the fan box and the cross plate at the rear of the Systems Cabinet.
8. Remove LOTO from the Systems Cabinet.

Testing the Magnet Monitoring Unit (MMU)

After servicing the MMU, perform the tests listed in the following table.

Table 2-8: MMU Tests

Test Description	Reference Document	Document #	Section
MMU	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Magnet Monitoring Unit
Quench	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	3 - System Installation / Cooling Down the Magnet
Modem dial-out	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Configuring MMU Remote Functions
Modem dial-in	<i>Optima 1.5T MR430s System Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Configuring MMU Remote Functions

Host Computer

Required Persons	Estimated Procedure Completion Time
1	2 hours

Replacing the Host Computer

This procedure describes the removal and replacement of the Host Computer component in the Optima 1.5T MR430s Systems Cabinet. See [Figure 2-1](#) on page 7 to identify the host computer.



WARNING

When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.



WARNING

Possible personal injury! Heavy Object - The Host Computer with box weighs approximately 55 pounds/25 Kg. Lifting this product requires assistance (mechanical or additional personnel).

To replace the host computer:

1. Complete the **procedure** [Preparing the Systems Cabinet for Service](#) on page 8.
2. Disconnect the **power cord** from the rear of the host computer.
3. Disconnect all the **cables** at the rear of the host computer.
4. Remove the **four screws** that secure the front of the host computer to the systems cabinet.
5. Slide the **host computer** out the front of the systems cabinet.
6. To install a replacement host computer, reverse the preceding steps, and make sure the **power switch** at the front of the host computer is switched ON.
7. Perform the necessary steps of the SW Reinstallation. Example of typical steps:
 - a. Preparing Hard disk
 - b. Loading Software
 - c. Installing Silhouette Licenses
 - d. Configuring Computer and Network Settings
 - e. Restoring Configuration Files
 - f. Restoring Magnet Shim File
 - g. Restoring system protocols
 - h. Loading Single Media Archive (SMA)
 - i. Loading DICOM Worklist

- j. Loading Operator Documentation
- k. Loading Webex
- l. Testing Section

Chiller

Required Persons	Estimated Procedure Completion Time
2	1 hours

Replacing the Chiller

This procedure describes the removal and replacement of the Chiller component in the Optima 1.5T MR430s Systems Cabinet.

Tools and Test Equipment	Quantity	Part Number
Bucket	1	2324086-2
Paper towels or rags	As needed	

⚠ WARNING When connected to building power, the systems cabinet contains potentially lethal voltages. Always complete the procedure [Preparing the Systems Cabinet for Service](#) before performing any service procedure inside the systems cabinet.

⚠ WARNING Possible personal injury! Heavy Object - The Chiller with box weighs approximately 104 pounds/47 Kg. Lifting this product requires assistance (mechanical or additional personnel).

To replace the chiller:

1. Complete the **procedure** [Preparing the Systems Cabinet for Service](#) on page 8.
2. Disconnect the cables and power cord from the rear of the Chiller.

Figure 2-25 Rear of Chiller



3. Disconnect the coolant hoses at the quick disconnect outside the Cabinet. Be sure that the quick disconnect shuts off both ends of each hose connection. Slide the hoses through the bulkhead panel after removing the rubber grommets. Do not drain the Chiller at this time. It will be easier when the Chiller is sitting on the Manual Lift after removal.

Figure 2-26 Chiller Quick Connects and Bulkhead Panel



- 4. Loosen the 4 screws that secure the Power Distribution Unit (PDU) to the front of the Systems Cabinet.

Figure 2-27 Power Distribution Unit



NOTICE *Dispose of any rags or used paper towels in accordance with local requirements.*

- 5. Remove the 8 screws that secure the Chiller to the front of the Systems Cabinet.

Figure 2-28 Front of Chiller



6. With the Manual Lift nearby and 1 person lifting the front of the PDU upwards, the other person shall slide out the Chiller from the Cabinet front. When the Chiller is about ½ removed from the Cabinet, secure the Manual Lift in front of the Chiller by lifting the platform up to the bottom height of the Chiller and applying the caster brakes.
7. Finish sliding out the Chiller from the Cabinet and onto the Manual Lift platform, unlock the caster brakes, and move Lift and Chiller out of the way.
8. Lift Chiller above a bucket to drain the coolant using the drain hose supplied with the new chiller.

Figure 2-29 Drain Hose Attachment

9. To install a replacement Chiller, set a new one on the platform and reverse the preceding steps. Fill replacement Chiller with coolant drained from the original chiller after installation.
10. Remove LOTO from the Systems Cabinet.



Testing the Chiller

1. Make sure the Chiller is powered up.
2. Perform a Daily QA on the system.
3. Check Chiller temperature about 15 minutes after the completion of the Daily QA. It should be 20°C.
4. Check that the pressure gauge reads between 60-80 psi.
5. Check the chiller connections to make sure there are no leaks.

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3 Operator Console



Replacement of any component with anything other than a genuine GE Field Replaceable Unit will void the regulatory certificates.

Power Control Unit (PCU) Replacement

Required Persons	Estimated Procedure Completion Time
1	10 minutes

This procedure describes the replacement of the Power Control Unit (PCU) of the Optima 1.5T MR430s System.

1. Perform the LOTO (Lock-out/Tag-out) procedure for the Systems Cabinet and to power-down the Power Control Unit (PCU).
2. Disconnect the power control cable from the rear of the PCU.
3. Connect the power control cable to the rear of the new PCU.
4. Remove LOTO from the Systems Cabinet.

Testing the PDU

Perform these tests with the power switches of all system cabinet components on, and all circuit breakers on the front of the PDU in the **On** position.

1. Verify that when the system is in Full On mode, all components are powered on.
2. Verify that when the system is in Standby mode, only the scanner computer, Digital IO, Exciter/Receiver, and MMU are powered on.
3. Log out of the applications software before executing the next step.
4. Verify the when the system is in Emergency Stop mode, all units in the cabinet are powered off. After an Emergency Stop you must go into Full on Mode to turn the system back on.

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4 Magnet Assembly

This section includes procedures for replacing the following FRU. Other components either are not field-replaceable, or do not require procedures.

- Coldhead

Replacing the Coldhead

Required Persons	Estimated Procedure Completion Time
2	8 hours

This procedure describes how to replace the 5001-0015 coldhead.

NOTICE

Replacing the coldhead exposes the helium space in the magnet to atmosphere. This dictates that the procedure be completed as quickly as possible, to prevent quenching the magnet. Plan your movements, and prepare the tools and materials needed in advance.

Materials Required

Description	Part Number	Comments
Vacuum grease	2297760	
Installation wrench kit	2000-9405	
Dewar of liquid helium		250 liters
Bottle of compressed helium gas		99.9999% - "5 nines"
Large soft cloth (to place chilled coldhead on)		
Sheet of clear rigid plastic large enough to cover the magnet bellows opening		
Ventilation fan		minimum 700cfm, 20m ³ /min
Transfill kit	2000-9401 or 2000-9411	
Ramp kit	2000-9403 or 2000-9413	
Non-magnetic tools	5112581	
Feeler gauge		
Anti seize		
Personal O ₂ monitor		
Torso/extremity protection		
Cryogenic gloves		
Earplugs		Grainger part number 4T151
Teslameter	46-251865 G2, G3, G4, G5	
Small sample kit	2000-9406	

Procedure

⚠ WARNING

Boil-off from cryogenic liquids can displace the breathable air in the scan room. To prevent possibly fatal asphyxiation, ensure that there is adequate ventilation, that the space is not confined, and that you do not remain in the area longer than necessary.

⚠ WARNING

Contact with cryogenic liquids can cause severe tissue damage. Always wear insulating gloves and eye protection when transferring cryogenic liquids.

NOTICE

Make all appropriate site notifications.

To replace the coldhead:

1. Examine the **replacement coldhead**, to see if it has the vibration dampers installed. If it does not, you will have to use the dampers from the old coldhead.
2. Determine whether the replacement coldhead package includes a **base O-ring**. If it does not, you will have to use the one from the old coldhead.
3. Ramp down the **magnet** (see [Ramping Down the Magnet](#) in the *OPTIMA MR430S System Installation Guide*).
4. If this is the third or fourth time you are replacing the coldhead, replace the adsorber (see [Replacing the Adsorber](#) on page 47).
5. Remove the **RF coil** from the magnet.
6. Lift the **upper cover** off the magnet, and place it out of the way.
7. Perform the LOTO procedure for the cryo-cooler compressor.
8. Disconnect the **helium supply and return elbows** from the coldhead.
9. Disconnect the **coldhead power cable** from the connector below the helium supply and return elbows.

Figure 4-1 Helium Supply and Return Elbows

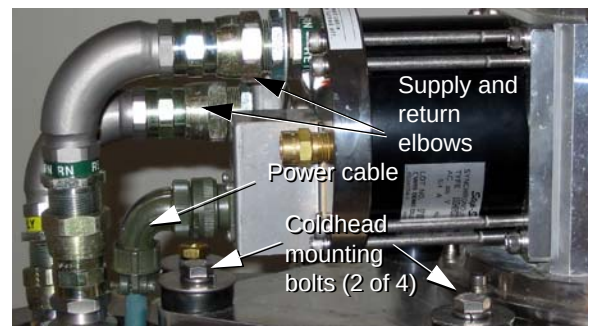
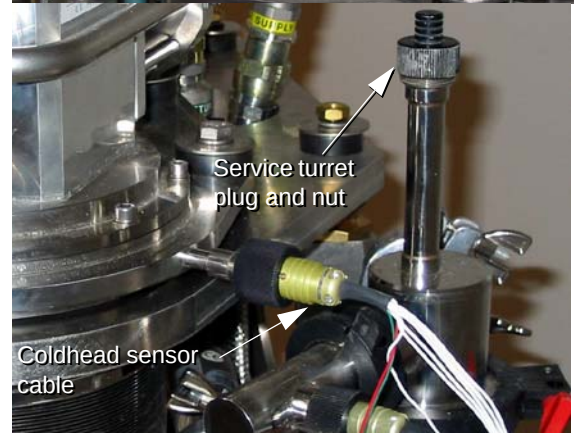


Figure 4-2 Sensor Cable and Turret Plug

10. Loosen the **nut** that secures the plug to the top of the service turret, to allow pressure from the helium boil-off to escape. If you cannot easily move the plug as you unscrew the nut, you have negative pressure in the magnet and you must wait for the magnet pressure to become positive before removing the nut and plug and proceeding to the next step.
11. Loosen the **four bolts** that secure the coldhead to the coldhead plate (see [figure 4-1](#) on page 34), and back them all the way out of the holes in the coldhead plate. Remove the bolts. The bolt underneath the coldhead power connector may be trapped by the connector; if so, leave it in the coldhead base plate until the coldhead is out.

**CAUTION**

Make sure the helium boil-off does not pressurize the space under the coldhead. Excess pressure under the coldhead can cause the coldhead to rise suddenly when the bolts are removed, possibly damaging the coldhead. You may have to remove the plug from the service turret and loosely stuff the opening with a paper towel to ensure that the boil-off can escape quickly enough.

12. Lift the **coldhead** straight up out of the bellows, being careful not to hit the side of the bellows. Place the coldhead on a large, soft cloth.

WARNING

The surfaces of the coldhead that were in the magnet bellows are extremely cold. To prevent injury, do not touch these surfaces with bare hands.

CAUTION

If the coldhead will not lift out of the bellows, it is probably frozen in place by ice. Let the coldhead warm for two minutes, and then try again using a twisting motion.



When the coldhead is removed, it begins to warm, and the relief valve lets off pressure with an explosive sound. Do not let this sound startle you and make you drop the coldhead.

13. Quickly cover the opening of the magnet bellows with a foam plug or plexiglass, to prevent ice crystals from falling into the magnet.



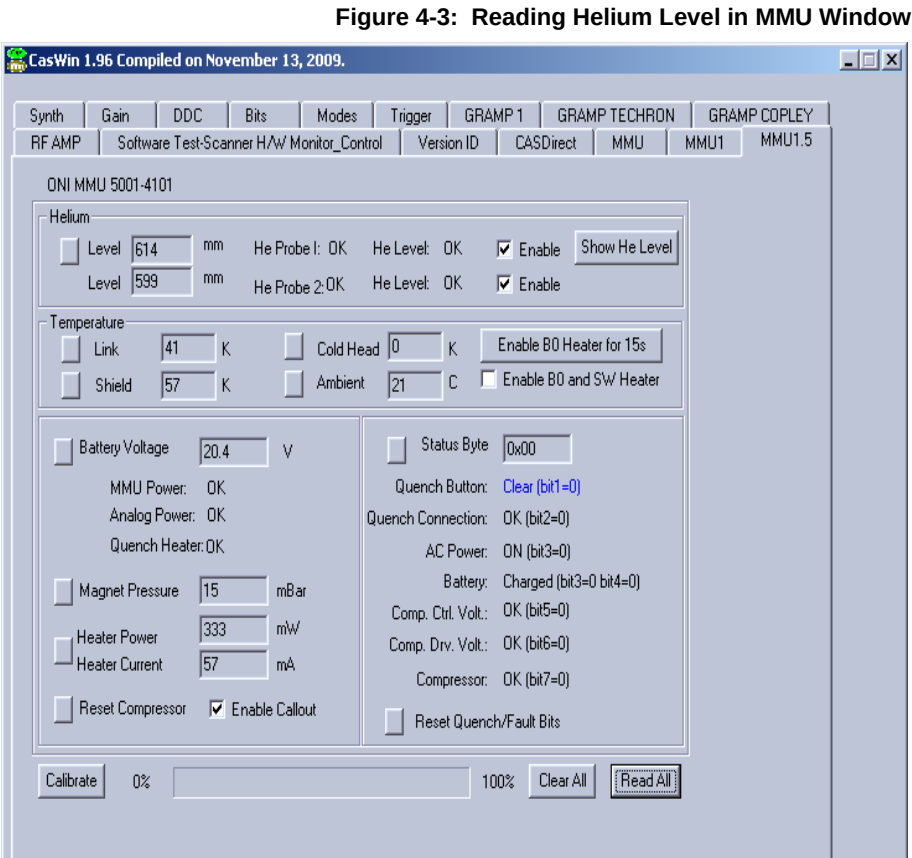
Never leave the coldhead turret uncovered for any longer than necessary.

14. Inspect the **bore** of the bellows through the clear plastic sheet. If there is ice in the bore, blow it out of the bore using compressed helium gas.
15. Remove the **fourth bolt** (under the coldhead power connector) and the **vibration dampers** and **O-ring**, if required, from the old coldhead.
16. Grease the **O-ring** and install it on the replacement coldhead.
17. Install the **vibration dampers** on the replacement coldhead.
18. Install the **bolt** under the coldhead power connector of the replacement coldhead.
19. Remove the foam plug or plexiglass covering the magnet bellows, and carefully lower the **replacement coldhead** into the bellows.
20. Install the four **bolts** that secure the coldhead to the coldhead plate.
21. Install the **service turret plug**, and tighten the nut.
22. Connect the **coldhead power cable**.
23. Connect the helium **supply and return elbows**.
24. Check the compressor **static pressure**, and recharge the compressor with five 9s helium if necessary.
25. Check the **helium level** in the magnet (see [Figure 4-3](#)).
 - a. Log in to the scanner application and enter **Service** mode.
 - b. Click **Service Tools**, then click **Free CAS**.

The system frees the CAS bus for use by CasWin.
 - c. Click **Service Tools**, then click **CasWin**.

The CasWin window opens.
 - d. Click the **MMU1.5** tab.

- e. Click the Helium Level button, and read the helium level.
26. Before ramping, ensure at least 600mm of helium. Fill the **magnet** with liquid helium if necessary. (See [Filling with Liquid Helium](#) in the *OPTIMA MR430s System Installation Guide*).
27. **Ramp** the magnet. (See [Ramping the Magnet](#) in the *OPTIMA MR430s System Installation Guide*).
28. Top off the magnet if necessary.
29. Verify that the **static pressure** shown on the gauge at the rear of the compressor is between 1.6 MPa and 1.65 MPa. If the pressure is below 1.6 MPa, charge the compressor.
30. Remove LOTO from the cryo-cooler compressor and start the **compressor and coldhead**.
31. Verify that the **dynamic pressure** on the gauge at the front of the compressor is between 1.9 and 2.2 MPa.



NOTICE

Immediately after changing the coldhead, the magnet pressure will be higher than normal. Monitor the magnet pressure and ensure that the pressure goes down and stabilizes at specification (see the PM Datasheet for specification values). If the pressure does not go down, you will need to remove any ice in the sleeve.

Testing the Coldhead

Table 4-1: Coldhead Tests

Description	Ref Doc	Doc #	Section
MMU	Optima 1.5T MR430s Installation Guide	DOC0763285	4 - System Config. and Test
Pressure Regulator	Optima 1.5T MR430s Installation Guide	DOC0763285	4 - System Config. and Test
Shim to specifications	Shimming Guide	DOC0797561	--
Center Frequency	Optima 1.5T MR430s Installation Guide	DOC0763285	4 - System Config. and Test
Small Sample/Stability	Optima 1.5T MR430s Installation Guide	DOC0763285	4 - System Config. and Test

5 Gradient Coil

This section includes the following sections:

- [Centering the Gradient Coil](#) on page 37
- [Gradient Coil Replacement](#) on page 38
- [Testing the Gradient Coil](#) on page 46

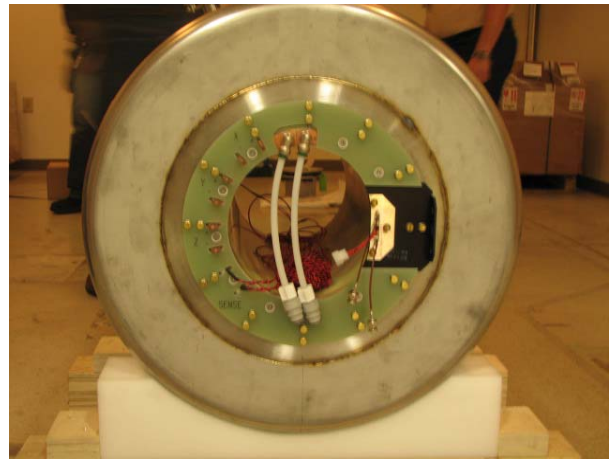
Centering the Gradient Coil

This procedure may be required when the coil is disturbed or replaced. The procedure requires plastic shim stock cut into 2-inch by 3-inch rectangles. The shim stock thicknesses are:

- 0.003 inch (0.0762 mm)
- 0.005 inch (0.127 mm)
- 0.010 inch (0.254 mm)

To center the gradient coil during the coil replacement:

Figure 5-1 Gradient Coil , Rear View



1. Pull the **gradient coil** out several inches from the service end of the magnet to allow easier shim insertion.

Figure 5-2 Gradient Coil , Front View

2. Create equal sized stacks of **shim stock**.
3. Insert the four **shim stacks** at four equidistant locations around the perimeter of the gradient coil.
The most likely positions are at 0°, 90°, 180°, and 270°.
4. Carefully push the **gradient coil** forward, wedging the four shim stacks securely.
5. Adjust the thickness and position of the **shim stacks** to center the gradient coil bore in the magnet.
6. Make sure the **shim stock** cards do not project past the edge of the magnet bore.



Gradient Coil Replacement

Required Persons	Estimated Procedure Completion Time
2	7 hours

This procedure describes the removal and replacement of the 5002-0000-002 Gradient Coil in the Optima 1.5T MR430s Magnet Unit.

**WARNING**

Possible personal injury! Heavy Object - The Gradient Coil weighs approximately 45 pounds/20 Kg. Lifting this product requires assistance (additional personnel).

Materials Required

Description	Part Number	Comments
Small sample kit	2000-9406	
Permanent marker		
Paper towels or rags		
Electrical tape		

Procedure

1. Perform the LOTO procedure for the Systems Cabinet.
2. Remove the RF Coil from the Magnet bore (see [Figure 5-3](#)).

Figure 5-3 Magnet Bore



3. Remove the Front Cover and carefully it set aside (take care to protect the finish). The cover is held on by Velcro strips. Be careful of the laser leads attached to the front cover (see [Figure 5-4](#)).

Figure 5-4 Front Cover Removal



⚠ WARNING Shim trays contain ferrous material.

4. Remove the Shim Trays and set outside the scan room as these will be used with the replacement gradient coil (see [Figure 5-5](#)).

Figure 5-5 Shim Trays



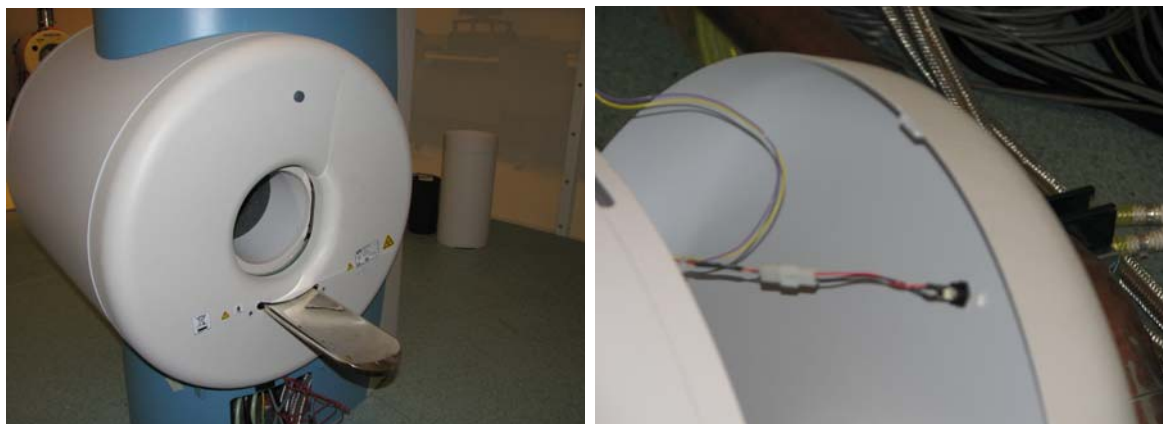
5. Remove the Utilities Cover at the bottom rear of the unit (see [Figure 5-6](#)).

Figure 5-6 Utilities Cover Removal



6. Remove the Rear Cover by removing the two Allen head screws attached to the extremity support bracket and by releasing the tab located at the top of the cover (see [Figure 5-7](#)).

Figure 5-7 Rear Cover Removal



7. Slide the Rear Cover off over the Extremity Support Bracket (see [Figure 5-8](#)).

Figure 5-8 Sliding the Rear Cover Over the Extremity Support Bracket



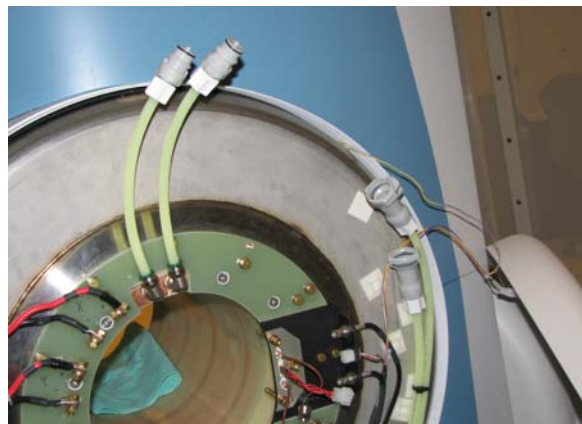
8. Remove the Extremity Support Bracket by removing the 2 screws and 2 bolts (see [Figure 5-9](#)).

Figure 5-9 Extremity Support Bracket Removal

**NOTICE**

Paper Towels will be needed in the next 2 steps for coolant leaks.

9. Disconnect the gradient cooling hoses from the rear of the Gradient Coil at the quick-disconnects (see [Figure 5-10](#)).

Figure 5-10 Disconnecting Cooling Hoses

10. At the front of the Gradient Coil, disconnect the Cooling Loops between the Shim Drum and the Gradient Coil. The outer sleeve of the tubing will need to be slid away from the gradient coil connection before the tubing can be removed. Replacement tubes are provided with the Gradient Coil, so remove the tubing from the shim drum as well (see [Figure 5-11](#)).

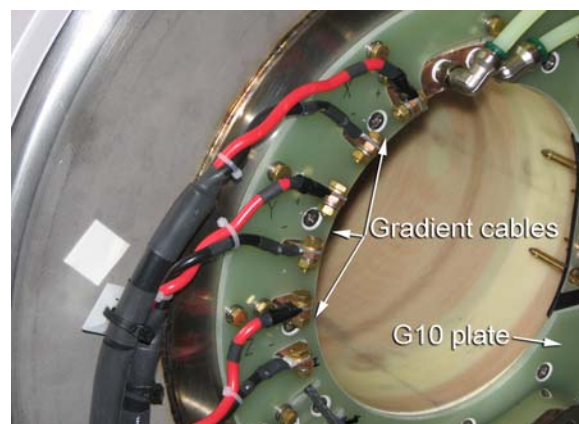
Figure 5-11 Shim Drum Coolant Tubing

11. Disconnect the Coil ID cable and the two RF cables from the RF Bracket at the rear of the Magnet (see [Figure 5-12](#)).

Figure 5-12 RF Bracket

12. Disconnect the 6 Gradient Cables from the Gradient Coil (see [Figure 5-13](#)).

Figure 5-13 Gradient Cables and G10 Plate



13. Remove the G10 Plate (see [Figure 5-13](#)).
14. Remove the RF Bracket (see [Figure 5-14](#))

Figure 5-14 RF Bracket Removal



15. Disconnect the two gradient sensor molex connectors P1 and P2. (If necessary, follow the black and red temp sensor cables from the back of the gradient coil to the molex connectors)
16. Bring the temperature sensor wires to the gradient coil and route as necessary to free up the sensor cables. (See [Figure 5-15](#))

NOTICE

The P2 connector is routed through the G10 plate and can not be freed without cutting the temperature sensors. If it becomes required to cut the temperature sensors, make sure the sensor wires are labeled. (see [Figure 5-16](#)).

17. Cut the two temperature sensors ~ 4 inches (100 mm) from the P2 connector and label the sensors. These will be spliced into the gradient coil temp sensor cable later.

Figure 5-15 Sensor Cable Wire Path

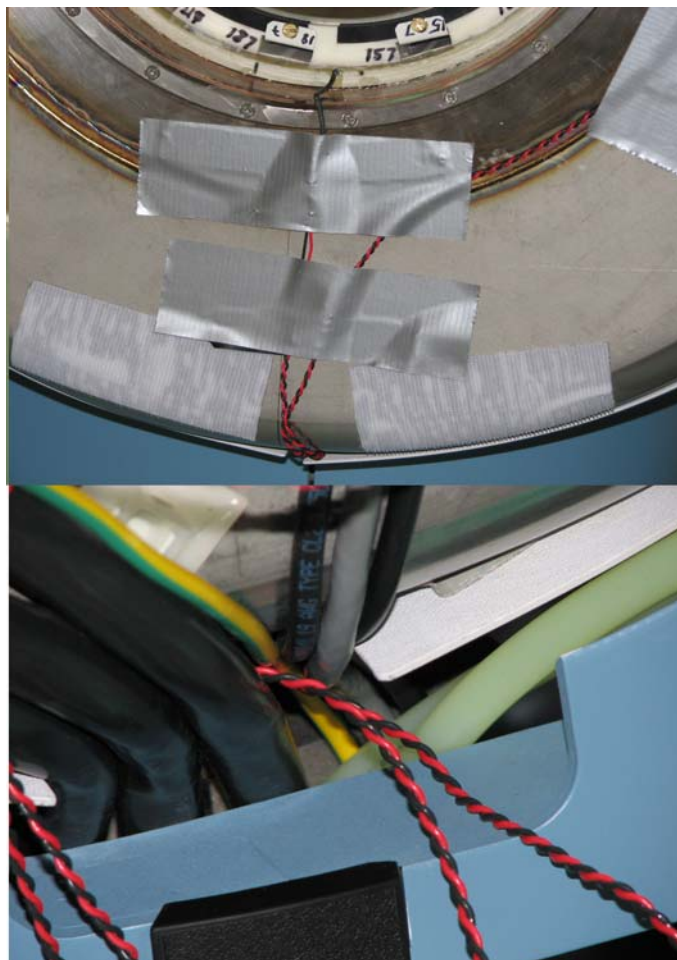


Figure 5-16 Sensor Wires



18. The Gradient Coil should now be free to be removed. Gradually and carefully slide the Coil out the rear of the Magnet and set it aside. Use one of the old pieces of tubing to connect the two front coolant parts.
19. Install the replacement Gradient Coil by sliding it through the Magnet bore.
20. Carefully route the 2 gradient coil temperature Sensor Cables to the rear of the Magnet.
21. Cut the two internal gradient coil temperature sensors ~ 4 inches (100 mm) from the P2 connector. Route through the G10 plate (see [Figure 5-16](#)).
22. If necessary, center the gradient coil by following the procedure [Centering the Gradient Coil](#) on page 37.
23. Loosely reattach the RF Bracket.
24. Loosely reattach the G10 Plate. (Be careful not to break any screws).

25. Slide in an RF Coil to check the RF bracket positioning. Once the RF Coil slides in smoothly, tighten the RF Bracket and G10 Plate connections (see [Figure 5-17](#)).

Figure 5-17 RF Coil Connector

26. Connect all of the Gradient Cables and ground wire to the Gradient Coil.
27. Connect the Coil ID cable and the two RF cables to the RF Bracket.
28. Connect the new Cooling Loops between the Gradient Coil and the Shim Drum. Slide the outer sleeves over the tubing ends after the tubing has been slid over the fittings. Attach to the Gradient Coil first to help prevent kinking of the hose during assembly (see [Figure 5-18](#)).



NOTICE

There is a difference in length between the 2 cooling loops. The tubing for the 10:00 position is longer than the 12:00 tubing.

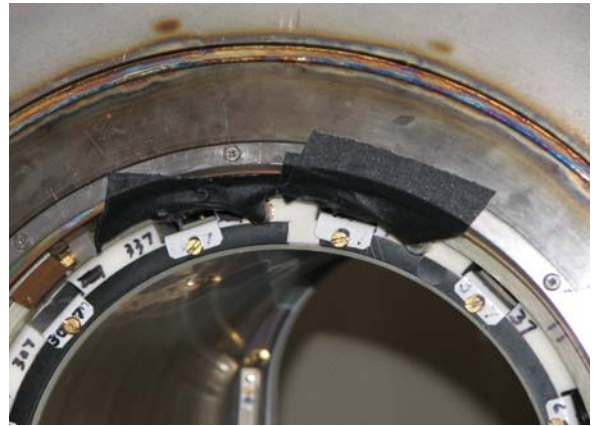
29. Relabel the gradient coil with the shim tray locations.
30. Connect the P1 molex connector.
31. Splice the two internal temperature sensors to the P2 molex connector. Wrap each splice connection with electrical tape. Reconnect the P2 molex connector.

Figure 5-18 Shim Drum Coolant Tubing



NOTICE

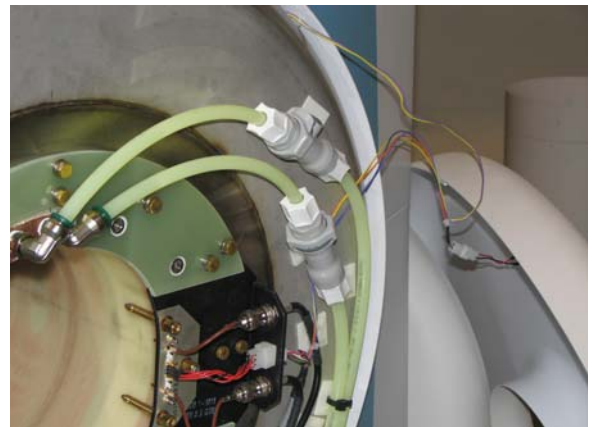
Cover the coolant lines with duct tape to prevent damage when inserting the shim trays see [Figure 5-19](#).

Figure 5-19 Coolant lines Covered with Duct Tape

32. Reconnect the quick-disconnects for the gradient cooling hoses (see [Figure 5-20](#)).

Figure 5-20 Reconnecting the Gradient Coolant Hoses

- 33. Reattach the Extremity Support Bracket.
- 34. Reattach the Rear Cover and Utilities cover.
- 35. Insert all of the original Shim Trays into the Gradient Coil.
- 36. Remove LOTO from system cabinet.
- 37. Check for leaks at the gradient coil coolant connections after the chiller restarts.
- 38. Replace Front Cover with laser on top at 2:00 position.

**NOTICE**

Drain the coolant from the failed gradient coil before returning. Dispose of any used rags or paper towels in accordance with local requirements.

Testing the Gradient Coil

After servicing the gradient coil or RF coil interface plate, perform the applicable test procedures listed in the following table.

Table 5-1: Gradient Coil Tests

Description	Reference Document	Doc #	Section
Gradient Calibration	<i>Optima 1.5T MR430s Installation Guide</i>	DOC0763285	4 - System Config. and Test
Eddy Current Compensation	<i>Eddy Current Compensation Procedure</i>	DOC0797560	
Gradient Calibration Recheck	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
Shim to specification	<i>Magnet Shim Test Procedure</i>	DOC0797561	--
Center Frequency	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
Small Sample/Stability	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
Image Quality/Ghosting	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
180mm RF coil SNR	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
123mm RF coil SNR	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test
Coil ID	<i>OPTIMA 1.5T MR430S INSTALLATION GUIDE</i>	DOC0763285	4 - System Config. and Test

6 Compressor

This section includes procedures for replacing the following components. Other components either are not field-replaceable, or do not require procedures.

- Adsorber (See [Replacing the Adsorber](#) on page 47)
- Fuses (See [Replacing Compressor Fuses](#) on page 47)

Replacing the Adsorber

Required Persons	Estimated Procedure Completion Time
1	1 hour

This procedure describes how to replace the adsorber in the helium compressor.

Equipment Required:

Description	Part Number	Comments
Replacement Adsorber		
Installation wrench kit	2000-9405	
Bottle of compressed helium gas		if compressor charge needed
Compressor change kit		if compressor charge needed
Appropriate material handling gloves		

Procedure

To replace the adsorber:

1. Log into the scanner application software and enter **Service Mode**.
2. Click **Service Tools**, then click **Free CAS**.
3. Start CasWin.exe from the Service Tools pull-down menu.
4. Click the MMU1.5 tab.
5. Deselect **Enable Callout**.
6. Perform the LOTO Procedure for the Cryo Cooler Compressor.
7. See section 3-1-1, Replacement of the Compressor Adsorber found in the CNA-31C or CNA-31D technical manual, DOC0798957 or DOC0798956.
8. Remove LOTO from the Cryo Cooler Compressor.
9. Re-enable the MMU Callout.
10. Verify that the static pressure shown on the gauge at the rear of the compressor is between 1.6 Mpa and 1.65 Mpa. If the pressure is below 1.6 Mpa, charge the compressor with five 9s helium.

Replacing Compressor Fuses

Required Persons	Estimated Procedure Completion Time
1	1 hour

There are three to six fuses inside the compressor.



Before replacing a blown fuse, correct the condition that caused it to blow. Install only the correct replacement fuse (see the [systems cabinet fuse parts list](#)).

To replace the fuses inside the compressor:

1. Log into the scanner application software and enter **Service Mode**.
2. Click **Service Tools**, then click **Free CAS**.
3. Start CasWin.exe from the Service Tools pull-down menu.
4. Click the **MMU1.5** tab.
5. Deselect **Enable Callout**.
6. Perform the LOTO Procedure for the Cryo Cooler Compressor.
7. See section 3-2, Fuse Replacement found in the CNA-31C or CNA-31D technical manual, DOC0798957 or DOC0798956.
8. Remove LOTO from the Cryo Cooler Compressor.
9. Re-enable the MMU Callout.
10. Verify that the static pressure shown on the gauge at the rear of the compressor is between 1.6 Mpa and 1.65 Mpa. If the pressure is below 1.6 Mpa, charge the compressor with five 9s helium.

Testing the Compressor

Table 6-1: Compressor Tests

Description	Reference Document	Document #	Section
MMU	<i>Optima 1.5T MR430s Installation Guide</i>	DOC0763285	4 - System Configuration and Test
Pressure Regulator	<i>Optima 1.5T MR430s Installation Guide</i>	DOC0763285	4 - System Configuration and Test

7 Patient Chair

This section contains procedures for replacing FRU components of the patient chair. Components not listed here either are not field-replaceable, or do not require procedures.

- Casters (See [Replacing the Casters](#) on page 49)
- Control handles (See [Replacing the Seat and Back Control Handles](#) on page 50)
- Hydraulic assemblies (See [Replacing the Hydraulic Assemblies](#) on page 51)
- Gas springs (See [Replacing Gas Springs](#) on page 56)

NOTICE

The right and left sides, and the front and rear of the chair in these procedures are from the viewpoint of a person sitting in the chair.

Replacing the Casters

This procedure describes how to replace the 2006-7701 casters.

Materials Required

Description	Part Number	Comments
Appropriate material handling gloves		

To replace casters:

1. Move the **chair** to an area large enough to lay it on its side.

CAUTION

Move the chair slowly when crossing door thresholds so as not to break off one of the stanchions from the base plate that holds the brake assembly. The chair must be replaced if this type of damage occurs.

2. Place foam, blankets, or other **cushioning material** on the floor next to the chair.
3. Tip the **chair** onto its side and lower it slowly onto the cushioning material.
4. Grasp the **caster**, and pull it straight out of the chair base.
5. To install a new caster, push its **stem** into the bore in the chair base.

Figure 7-1 Chair Laid Down



Figure 7-2 Removing Caster



NOTICE

The new caster stem may have mold flash that prevents it from pressing into the chair base. Remove this flash by sanding or filing the stem until it fits snugly in the bore of the chair base. Do not remove so much material that the caster comes out easily; it must be a snug fit in the base.

If the caster is a loose fit in the bore, apply RTV adhesive (P/N 46-170137P1) to the stem before inserting it in the bore. Let the RTV cure with the chair's weight on the casters for 24 hours before moving the chair across any thresholds.

Replacing the Seat and Back Control Handles

This procedure describes how to replace the 2006-1006 seat handles and how to install the 1091-0003-001, 1091-0003-002, 1091-0004-001, 1091-0004-002 labels.

To replace a control handle:

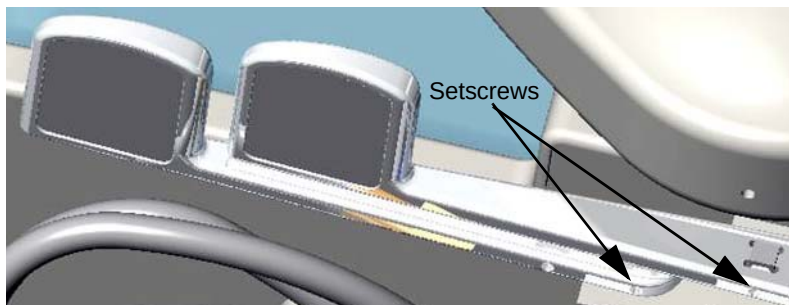
1. Remove the **setscrew** (1040-0002) from the control handle.

NOTICE

When replacing a forward control handle, the rear handle must be removed for access.

2. Slide the control **handle** off the square control shaft.
3. Slide the replacement **handle** onto the square shaft.
4. Apply a small amount of Loctite 222 to the **setscrew** and thread the setscrew through the handle and into the square shaft. Wipe off any excess Loctite.

Figure 7-3 Control Handle Setscrews



5. Install the appropriate **label** on the top of the handle.

NOTICE

Figure 7-4 at right shows the labels on the right-hand side of the chair. The labels on the left-hand side are similar.

Figure 7-4 Control Handle Labels



Replacing the Hydraulic Assemblies

Replacing the Upper Hydraulic Assembly

This procedure describes how to replace the 2006-0009 upper hydraulic assembly.

To replace the lower hydraulic assembly:

Release the Valve

1. Tilt the **seat back** to the rear.
2. Extend the **leg support**.
3. Remove the two screws that secure the **right-side lower cover** to the bottom of the seat, and remove the cover.
4. Remove the two screws that secure the **left-side lower cover** to the bottom of the seat, and remove the cover.
5. Remove the two screws that secure the **seat pad** to the chair, and remove the seat pad.
6. Raise the **seat back** to the upright position.

NOTICE

Raising the seat back to the upright position unloads the gas springs. You cannot disassemble the hydraulic assemblies with the gas springs loaded.

Figure 7-5 Upper Hydraulic Valve and Linkage

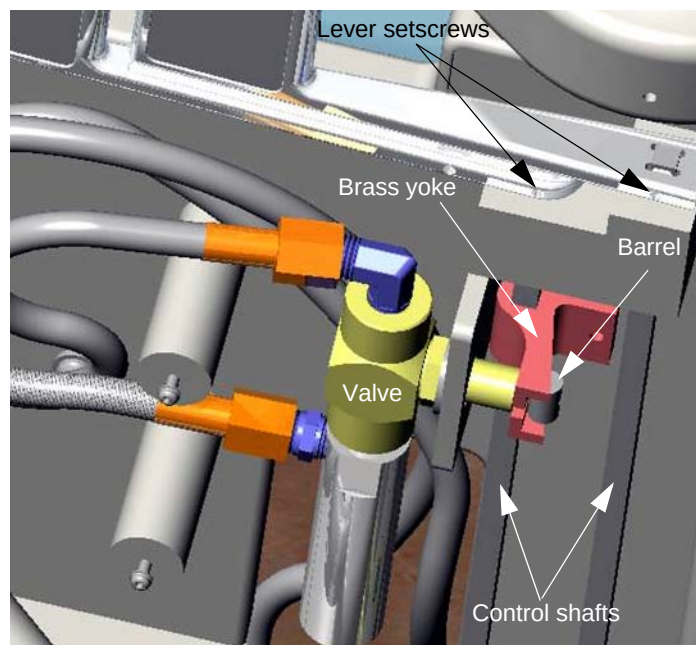
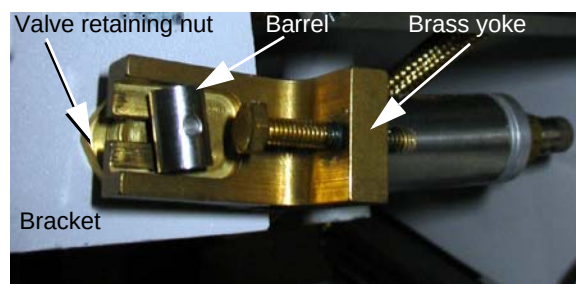


Figure 7-6 Removing Arm E-ring



7. Remove the E-rings that secure the **armrests** to the chair, and remove the armrests.
8. Remove the two **setscrews** that secure the left-side seat control levers to their shafts. (See Figure 7-5.)
9. Remove the seat **control levers**.
10. Slide the two square seat **control shafts** toward the right side of the chair until they are clear of the brass yoke. (See Figure 7-5.)

Figure 7-7 Valve Linkage

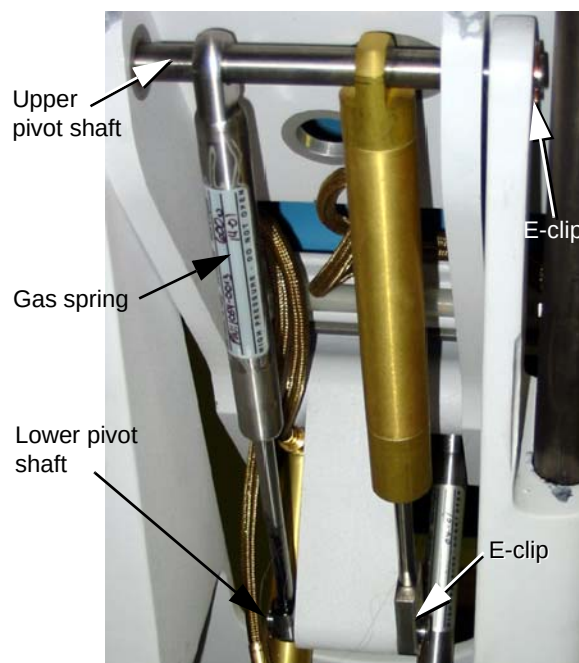


11. Remove the **brass yoke**.
12. Unscrew the **barrel** from the end of the valve shaft.
13. Remove the valve **retaining nut** from the valve stem.
14. Slide the **valve stem** through the bracket, and remove the washer from the valve stem.

Release the Hydraulic Cylinder

1. Raise the leg support.
2. Remove the E-clip from the left-hand end of the lower front pivot shaft, and slide the lower end of the hydraulic cylinder off the shaft.
Right and left are as seen when sitting in the chair.
3. Remove the E-clip from the left-hand end of the upper front pivot shaft.
4. Rock the seat base gently while pushing the shaft to the right, until the upper end of the hydraulic cylinder can slide off the shaft.

Figure 7-8 Upper Hydraulic Cylinder Attachment



Do not push the shaft with steel or other hard materials. Hard materials can damage the shaft.

NOTICE

While pushing the shaft, keep the gas spring from moving with the shaft. If the spring moves with the shaft, it can bind the shaft. A spacer of some kind placed between the spring end and the shaft bracket can help.

Figure 7-9 Passing Cylinder Up Through Seat Base

5. Pass the upper end of the hydraulic cylinder up through the center opening in the seat base, as shown at right.



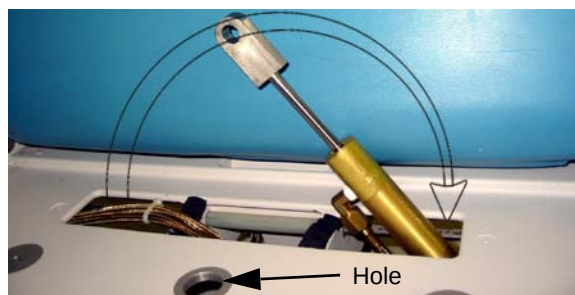
Figure 7-10 Passing Cylinder Down Through Seat Base

6. Rotate the cylinder end-for-end into the left-side opening in the seat base, and pass the cylinder down through the seat base.
Right and left are as seen when sitting in the chair.
7. Remove the lower hydraulic assembly from the chair.

Installing the Upper Hydraulic Assembly

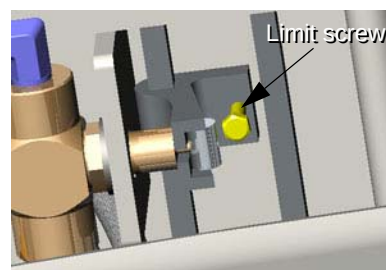
To install a new hydraulic assembly, reverse the preceding procedures, with the following considerations:

- Make sure the washer is installed on the valve stem, or the levers will droop.
- When threading the barrel on the valve shaft, the counterbore in the barrel is toward the valve.



- Verify that a **limit screw** is installed with Loctite on the valve control mechanism for the upper hydraulic assembly, and that the limit screw contacts the underside of the seat base before the valve stem is fully extended.

Figure 7-11 Limit Screw



You must apply Loctite to the limit screw. Without Loctite, the screw can work loose, and the control linkage can be damaged.

- When installing the seat assembly, align the hole in the bottom of the seat assembly with the hole in the middle of the seat base. (See Figure 7-10.)



To prevent patient injury, make sure that the E-clips are installed on the pivot shafts (at the upper and lower ends of the hydraulic cylinder; see figure 7-8 on page 52). Without all clips in place, the cylinder or shaft can work loose, allowing the seat back to fall.

Replacing the Lower Hydraulic Assembly

This procedure describes how to replace the 2006-0008 lower hydraulic assembly. The lower hydraulic assembly is under the right-hand side of the seat.

To replace the lower hydraulic assembly:

Release the Valve

1. Tilt the **seat back** to the rear.
2. Extend the **leg support**.
3. Remove the two screws that secure the **right-side lower cover** to the bottom of the seat, and remove the cover.
4. Remove the two screws that secure the **left-side lower cover** to the bottom of the seat, and remove the cover.
5. Remove the two screws that secure the **seat pad** to the chair, and remove the seat pad.
6. Raise the **seat back** and seat base to the upright position.

NOTICE

Raising the seat back to the upright position unloads the gas springs. You cannot disassemble the hydraulic assemblies with the gas springs loaded.

Figure 7-12 Lower Hydraulic Assembly

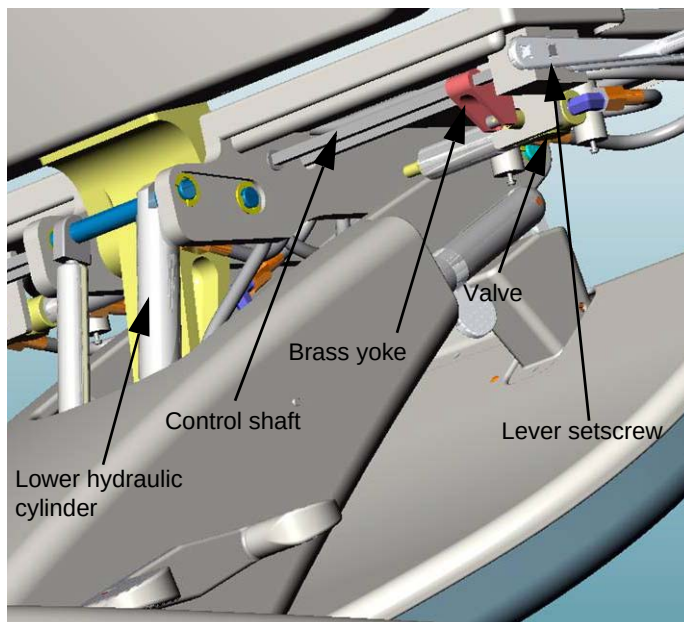


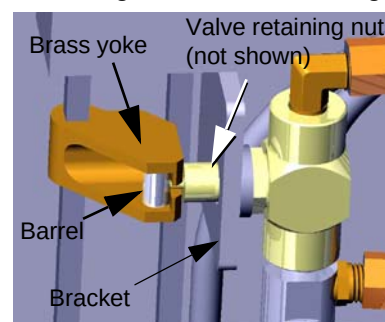
Figure 7-13 Removing Arm E-ring

7. Remove the E-rings that secure the **arms** to the chair, and remove the arms.
8. Remove the **setscrew** that secures the right-side seat back control lever to its shaft. (See Figure 7-12.)
9. Remove the seat back **control lever**.
10. Slide the square seat back **control shaft** toward the left side of the chair until it is clear of the brass yoke. (See Figure 7-12.)



11. Remove the **brass yoke**.
12. Unscrew the **barrel** from the end of the valve shaft.
13. Remove the valve **retaining nut** from the valve stem.
14. Slide the **valve stem** through the bracket, and remove the two washers from the valve stem.

Figure 7-14 Valve Linkage



Release the Hydraulic Cylinder

1. At the bottom front of the pedestal, remove the E-clip from the right-hand end of the lower rear pivot shaft, and slide the lower end of the hydraulic cylinder off the shaft.

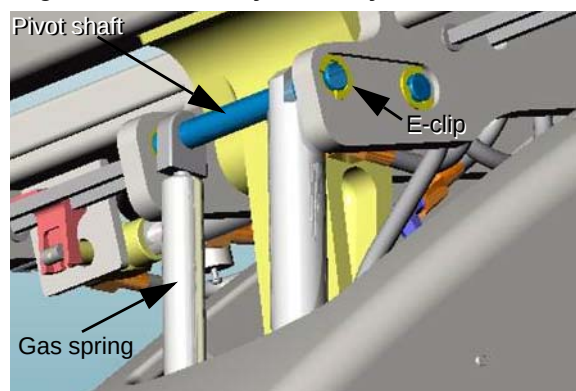
Right and left are as seen when sitting in the chair.

Figure 7-15 Lower Hydraulic Cylinder Lower Mount



Figure 7-16 Lower Hydraulic Cylinder Attachment

2. Remove the E-clip from the right-hand end of the upper rear pivot shaft.
3. Push the shaft to the left until the upper end of the hydraulic cylinder can slide off the shaft.



Do not push the shaft with steel or other hard materials. Hard materials can damage the shaft.

NOTICE

While pushing the shaft, stop the gas spring from moving with the shaft. If the spring moves with the shaft, it can bind the shaft. A spacer of some kind placed between the spring end and the shaft bracket can help.

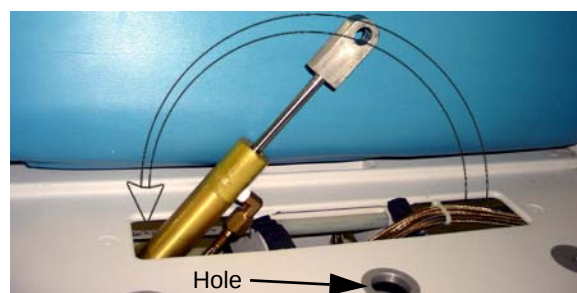
4. Pass the upper end of the hydraulic cylinder up through the center opening in the seat base.

Figure 7-17 Passing Cylinder Up Through Seat Base

5. Rotate the cylinder end-for-end into the right-side opening in the seat base, and pass the cylinder down through the seat base.

Right and left are as seen when sitting in the chair.

6. Remove the lower hydraulic assembly from the chair.

Figure 7-18 Passing Cylinder Down Through Seat Base

Installing the Lower Hydraulic Assembly

To install a new hydraulic assembly, reverse the preceding procedures, with the following considerations:

- Make sure the two washers are installed on the valve stem, or the levers will droop.
- When threading the barrel on the valve shaft, the counterbore in the barrel is toward the valve.
- When installing the seat assembly, align the hole in the bottom of the seat assembly with the hole in the middle of the seat base. (See Figure 7-18.)

⚠ WARNING

To prevent patient injury, make sure that the E-clips are installed on the pivot shafts (at the upper and lower ends of the hydraulic cylinder; see [figure 7-15](#) and [Figure 7-16](#) on page 54). Without all clips in place, the cylinder or shaft can work loose, allowing the seat to fall.

Replacing Gas Springs

Replacing the Upper Gas Spring

To replace the P/N 1054-0013 upper gas spring:

1. At the lower front of the chair pedestal, remove the E-clip that secures the shaft end of the gas spring to the lower pivot pin. Remove the shaft end of the spring from the pivot pin.
2. Unscrew the body of the gas spring from the eye on the upper pivot shaft. (See the figure at right.)
3. Unscrew the eye from the replacement gas spring.
4. Screw the replacement spring into the eye on the upper pivot shaft.

NOTICE

The upper and lower gas springs appear identical, but have different spring pressures. Make sure the replacement spring is part number 1054-0013. (The spring is marked with the manufacturer's number: GS3132-1.)

5. Slide the shaft end of the spring onto the lower pivot pin.
6. Install the E-clip on the end of the pivot pin.

Figure 7-19 Upper Gas Spring



WARNING

To prevent patient injury, make sure that the E-clip is installed on the pivot shaft (at the lower end of the gas spring; see [figure 7-19](#)). Without all clips in place, the hydraulic cylinder or shaft can work loose, allowing the seat back to fall.

Replacing the Lower Gas Spring

To replace the P/N 1054-0000 lower gas spring:

1. At the lower front of the chair pedestal, remove the E-clip that secures the shaft end of the gas spring to the lower pivot pin. Remove the shaft end of the spring from the pivot pin.

Figure 7-20 Lower Gas Spring



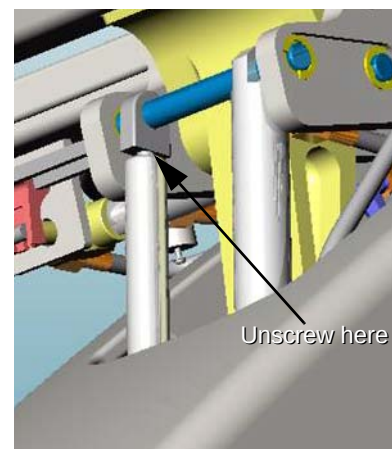
2. Unscrew the body of the gas spring from the eye on the upper pivot shaft at the rear of the pedestal. (See the figure at right.)
3. Unscrew the eye from the replacement gas spring.
4. Screw the replacement spring into the eye on the upper pivot shaft.

NOTICE

The upper and lower gas springs appear identical, but have different spring pressures. Make sure the replacement spring is the part number 1054-0000. (The spring is marked with the manufacturer's number: GS2433-1.)

5. Slide the shaft end of the spring onto the lower pivot pin.
6. Install the E-clip on the end of the pivot pin.

Figure 7-21 Upper End of Gas Spring



WARNING

To prevent patient injury, make sure that the E-clip is installed on the pivot shaft (at lower end of the gas spring; see [figure 7-20](#) on page 57). Without all clips in place, the hydraulic cylinder or shaft can work loose, allowing the seat to fall.

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8 Miscellaneous

Penetration Panel (Pen Panel)

Replacing the Pen Panel

- 1. Perform the LOTO procedure for the systems cabinet.
- 2. Replace the Pen Panel.
- 3. Remove LOTO.

Testing the Pen Panel

Table 1: Pen Panel Tests

Test Description	Reference Document	Document #	Section
Gradient Calibration Orientation	<i>Optima 1.5T MR430s Installation Guide</i>	DOC0763285	4 - System Configuration and Test
RFI	<i>Optima 1.5T MR430s Installation Guide</i>	DOC0763285	4 - System Configuration and Test / Miscellaneous Tests

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